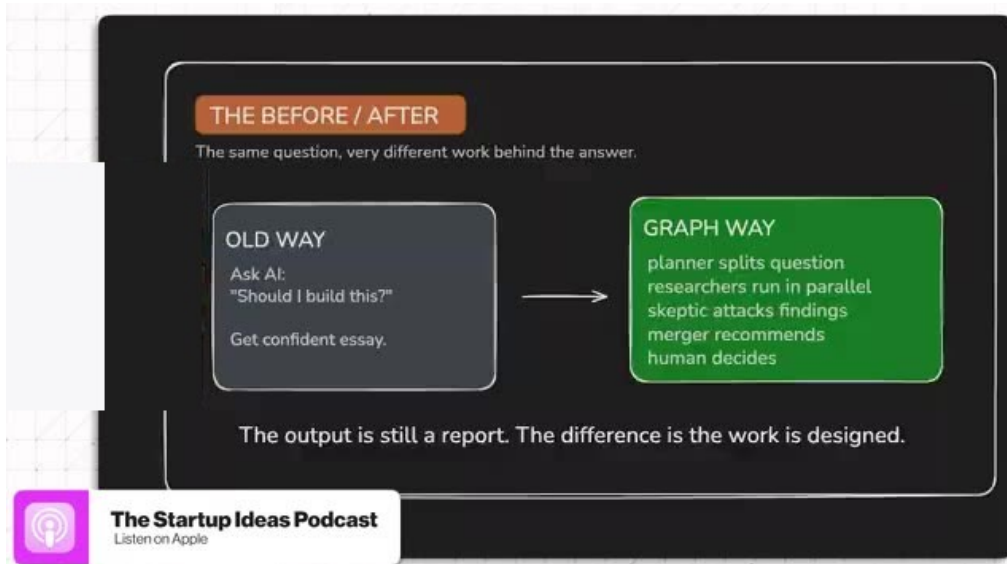


Designing AI Workflows That Can Be Trusted

The graph engineering pattern: jobs, arrows, and shared state — and when it earns its place

One model, one pass — no way to check the work



BUSINESS IMPLICATION

A single confident pass leaves no seam to check what evidence was used. Jobs, arrows, and shared state make the same work checkable at every step, not just at the end.

Decision: run the six-condition test (slide 8) before scaling risky AI work — then start with a manual first rep, not a tooling purchase.

The vocabulary is having a very contested moment

48 hrs

from first post to three competing definitions

3

incompatible meanings of “graph” in circulation at once

1

widely shared founding study — confirmed fabricated

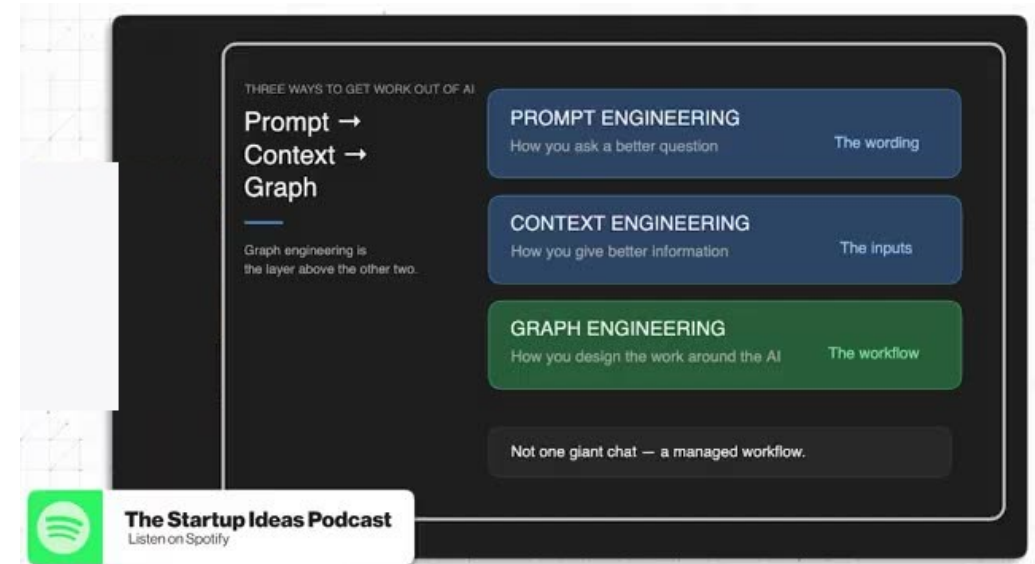
The term is new and contested. The mechanics underneath it are not — treat the label with more skepticism than the pattern.

Source: independent industry reporting, July 2026 (see run sources)

One active framing — not a settled progression

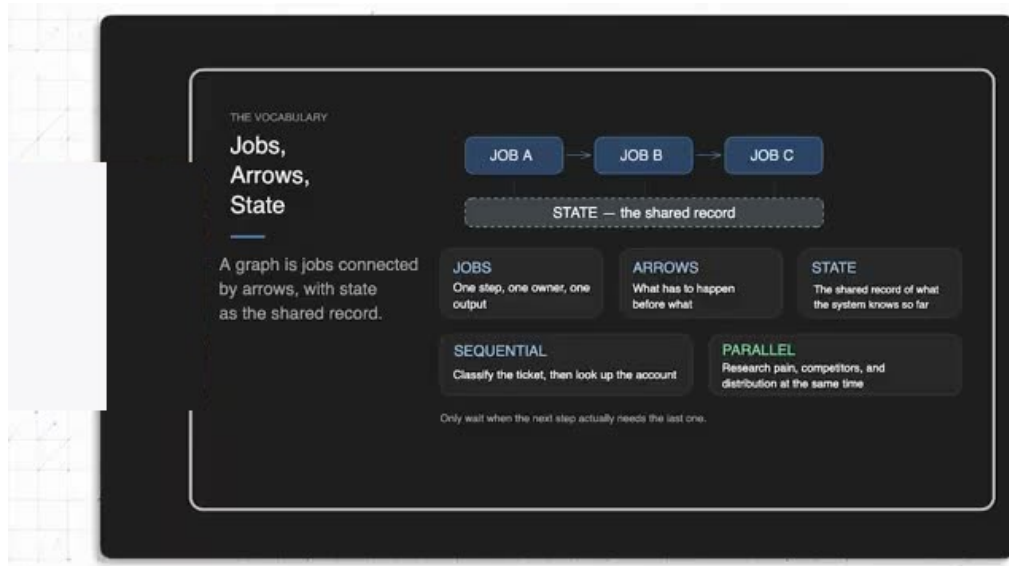
BUSINESS IMPLICATION

Prompt engineering is about wording, context engineering about information, graph engineering about how the work is designed — three levers, not an agreed sequence.



Take the mechanics (design the work, not just the prompt) without adopting the trend-cycle framing wholesale.

The vocabulary: jobs, arrows, and one shared record



BUSINESS IMPLICATION

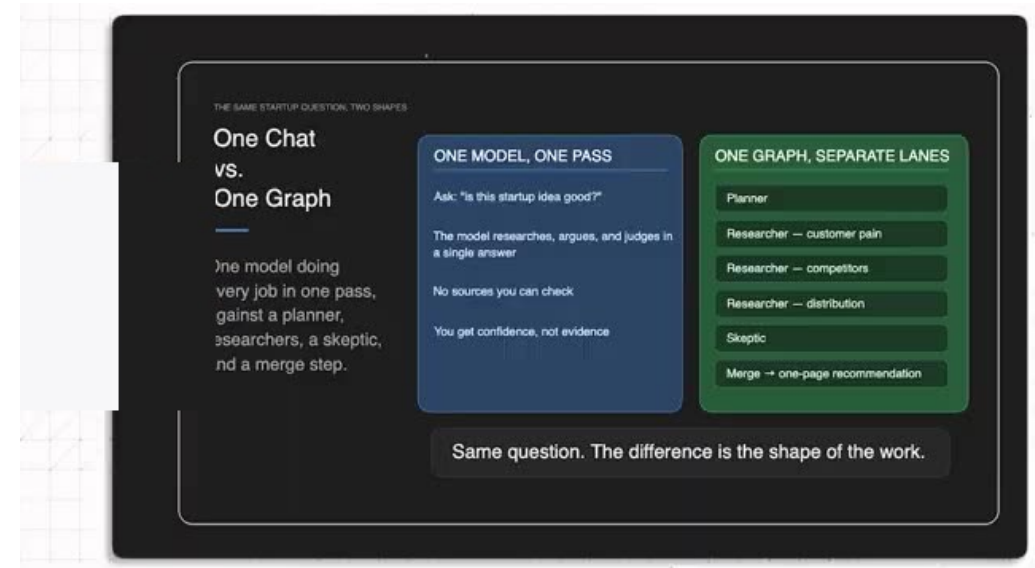
A job is one step with one owner and one output. An arrow says what has to happen before what. State is the shared record every job reads and writes. That's the entire vocabulary — nothing more exotic is required to start.

This is the whole mental model. Everything else in this deck is this vocabulary applied to a real decision.

Same report format — different shape of work behind it

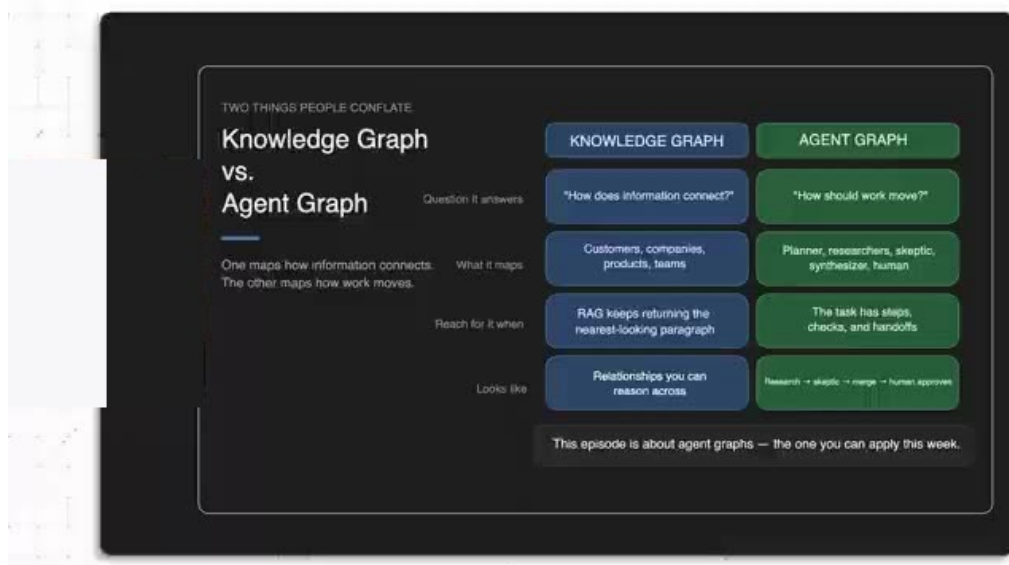
BUSINESS IMPLICATION

A single chat and a structured graph can both hand back a report. Only one of them lets you see which claims were checked, by whom, before you trust the answer.



Ask of any AI-produced recommendation: can I see the work behind it, or only the output?

Don't confuse a knowledge graph with an agent graph



BUSINESS IMPLICATION

A knowledge graph maps what your data means — customers, products, relationships. An agent graph maps how work should move, get checked, and get approved. This deck is about the second one.

Buying or building a knowledge graph does not give you a checkable AI workflow — they solve different problems.

The qualifying test: when a graph earns its place

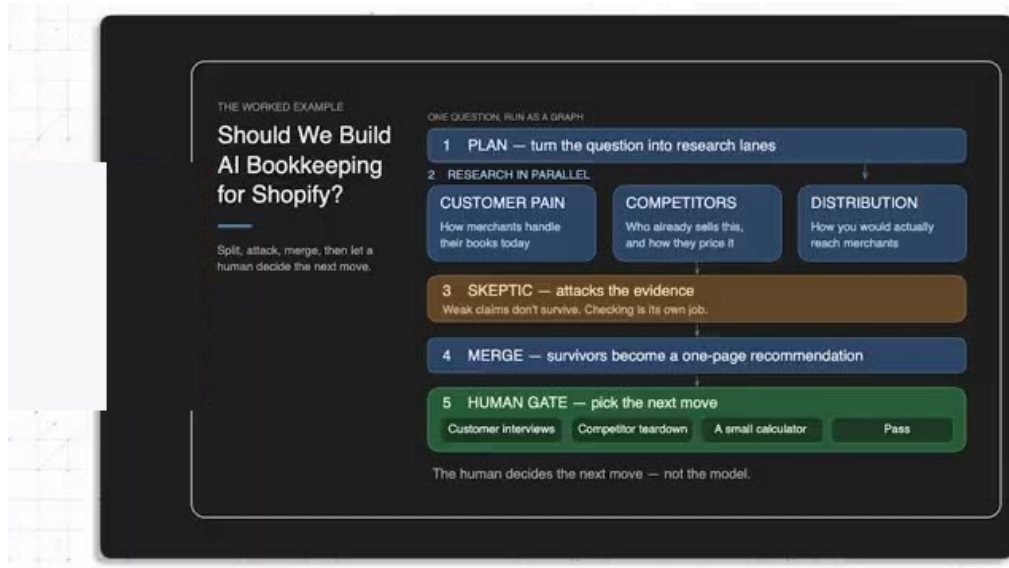
BUSINESS IMPLICATION

Multiple steps, multiple sources, parallel paths, real checks, real risk, required approvals — if none of these are true for a workflow, a graph is overhead, not sophistication.



Run every candidate workflow through this test before reaching for any framework. Most AI work still doesn't need one.

Worked example: AI bookkeeping for Shopify?



BUSINESS IMPLICATION

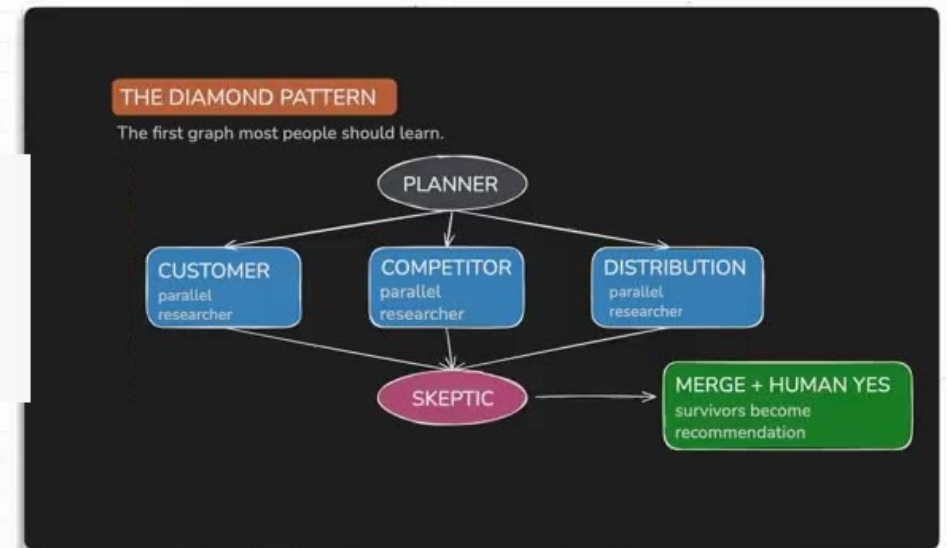
Three parallel researchers cover customer pain, competitors, and distribution. A skeptic attacks the findings. A merge step turns survivors into a one-page recommendation. A human picks the next move.

The messy one-shot chat becomes a workflow you can actually inspect, challenge, and reuse on the next decision.

The pattern underneath: the Diamond

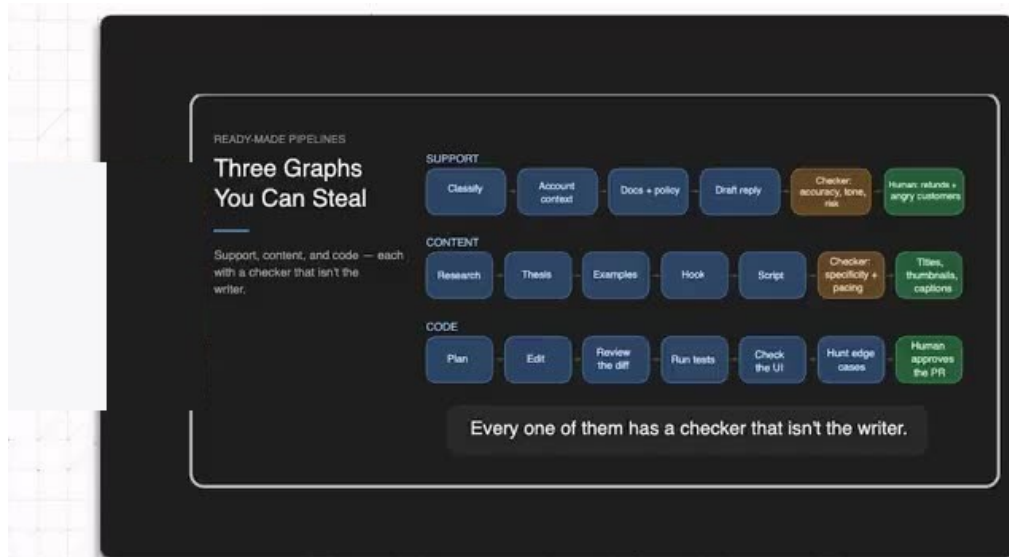
BUSINESS IMPLICATION

Planner fans out to parallel researchers, a skeptic checks their findings, then merge and human approval close the loop. This is the one shape worth learning first — every other pattern is a variation on it.



Learn this one diagram before anything else. It generalizes to support, content, and code work (next slide).

Three ready-made pipelines you can steal



BUSINESS IMPLICATION

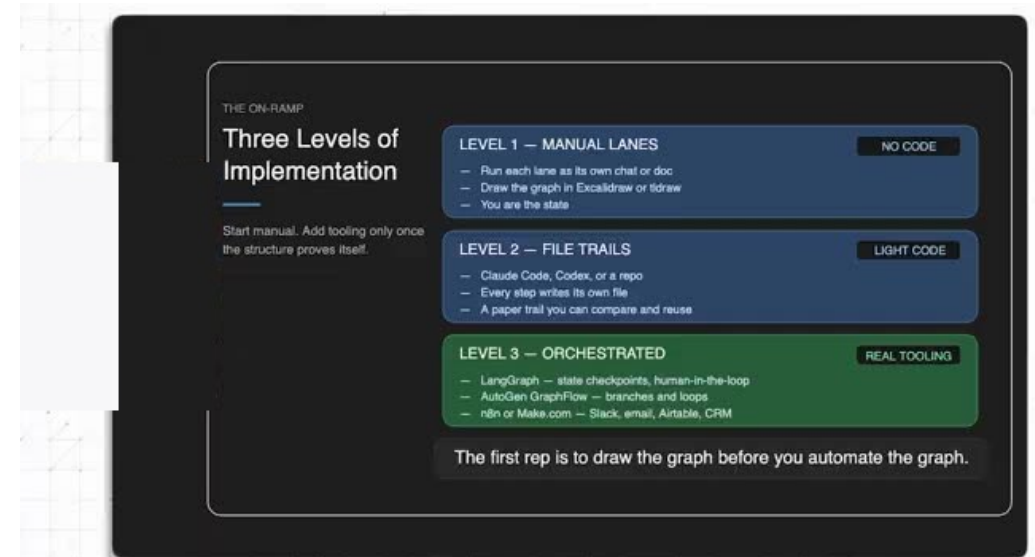
Support, content, and code workflows all follow the same rule: the checker is never the same role as the writer. That one constraint is what makes each pipeline trustworthy enough to run unattended.

Pick the lane closest to your own work and copy the structure — not the specific steps — into your first graph.

Three levels of implementation — start manual

BUSINESS IMPLICATION

Manual whiteboard lanes prove the shape works with no code at all. File trails make the run reusable. Real orchestration tooling only pays off after that — earn the tooling, don't start with it.



Do not buy or install anything before finishing Level 1 by hand at least once.

Where the named tools actually fit

Tool	Best for	Watch for
LangGraph	Stateful multi-agent logic, human-approval gates	Needs engineering capacity
AutoGen GraphFlow	Parallel fan-out, conditional/looping flows	Code-first, Python
n8n	Self-hostable, AI-native automation	Needs comfort with JSON/APIs
Make.com	Fully-hosted, non-technical automation	Billing punishes looping agents

Most teams should start below all four — see “Three levels of implementation.” Source: industry reporting, July 2026.

Next move: draw one graph this week

- 1 Pick one workflow you already run.
- 2 Draw the jobs and the arrows between them.
- 3 Delete the fake sequential waiting — run independent jobs in parallel.
- 4 Add one skeptic step before anything ships.
- 5 Run it once, by hand, before automating anything.